

COURSE DESCRIPTION

Course Number	0418 - 390	Course Title	Software Engineering
Credit Hours	3 (3:0:3)	Course Coordinator	Dr. Hazem Raafat
Compulsory	Yes	Prerequisite	0418-201, 0418-223

Textbook

Title:	Software Engineering: A Practitioner's Approach.
Author:	Roger Pressman, Bruce Maxim.
Publisher:	McGraw Hill, Latest Edition.

References:

Title:	Learning UML 2.0.
Author:	Russ Miles and Kim Hamilton.
Publisher:	O'Reilly Media, 2006.

Major Topics Covered in the Course

	Topics	Hours
1	Introduction to software engineering and processes	3
2	Classical and generic software development processes	3
3	Modern and specialized software development processes	3
4	Requirements engineering	3
5	Prototyping and formal requirements specification	3
6	Software architecture	2
7	Software design	3
8	Design of special systems (e.g., service-oriented systems)	2
9	Software modeling languages and tools	2
10	Software evolution	1
11	Fault tolerance, reliability, robustness, and security	3
12	Software configuration and release management	3
13	Verification and validation, software testing, and quality control	4
14	Software project management concepts and risk management	3
15	Ethical and professional issues	1
16	Exams, presentations, case studies	3
	Total.	42

Grading

20-30%	Midterm(s)
30-40%	Hmk/quiz/project
40%	Final

Course Outcome

Students should be able to:

1. Demonstrate mastering of software engineering principles.
2. Learn to analyze and design efficient and high-quality software for specific domains and computing environments.
3. Understand and follow appropriate software development processes.
4. Demonstrate knowledge and skills in verification and validation, software testing, and quality control.
5. Learn to use software development tools such as modeling tools, revision and code control tools, and IDEs.
6. Develop project plans and identify project risks.
7. Learning to efficiently and professionally perform software development team tasks such as preparing reports, giving technical presentations, and effective team collaboration.
8. Demonstrate knowledge of ethical, social, and professional issues related to software development.
9. Recognition of the need for and ability to independently do research on topics in software engineering techniques and tools.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1						H
2	H	H				H
3						H
4	L	L				H
5						H
6						H
7			H		H	
8				H		
9						H